ECS 409/609
Assignment 6
Sequential Construct-II
Floating Point Instructions, Input Output Instructions and Comparison Instructions

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Submission Deadline: 26 October, 2025

For code repository of this assignment: **MIPS Assignments**

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1. Problem 1

Write an MIPS assembly program that takes user rst name as an input and prompt the message

```
Hi Ajay, MIPS assembly programming is very exciting to learn
```

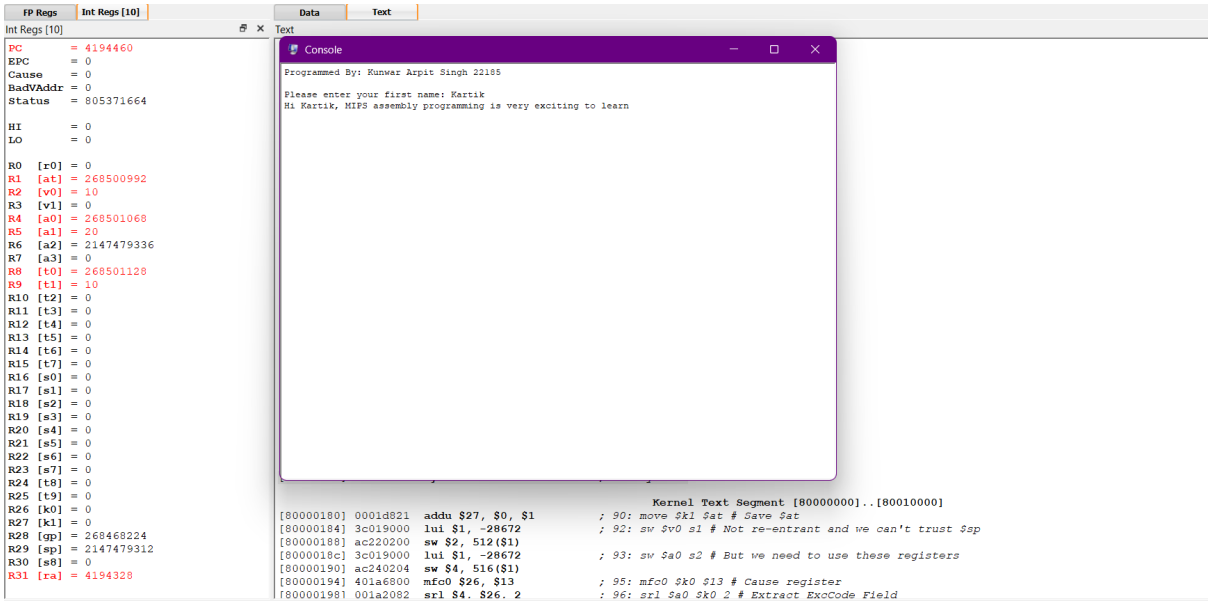
MIPS Assembly Source

```
# Assignment 6: Sequential Construct-II
# Programmed By : Kunwar Arpit Singh

.data
    prompt_msg: .asciiz "Programmed␣By:␣Kunwar␣Arpit␣Singh␣22185\n\nPlease␣
        enter␣your␣first␣name:␣"
    greeting_start: .asciiz "Hi␣"
    greeting_end: .asciiz ",␣MIPS␣assembly␣programming␣is␣very␣exciting␣to␣
        learn"
    user_name: .space 20

.text
.globl main

main:
    li $v0, 4
    la $a0, prompt_msg
    syscall

    li $v0, 8
    la $a0, user_name
    li $a1, 20
    syscall

    la $t0, user_name

find_newline:
```

```
    lb $t1, 0($t0)

    beq $t1, 10, replace_newline
    beq $t1, 0, exit_loop

    addi $t0, $t0, 1
    j find_newline

replace_newline:
    sb $zero, 0($t0)

exit_loop:

    li $v0, 4
    la $a0, greeting_start
    syscall

    li $v0, 4
    la $a0, user_name
    syscall

    li $v0, 4
    la $a0, greeting_end
    syscall

    li $v0, 10
    syscall
```

Output Screenshots

Register values and Console



2. Problem 2

Write an assembly program that takes principle amount, rate of interest and time (taken as an input to integer register) as an input from the user and calculate simple interest and display the raw and absolute results to the user.

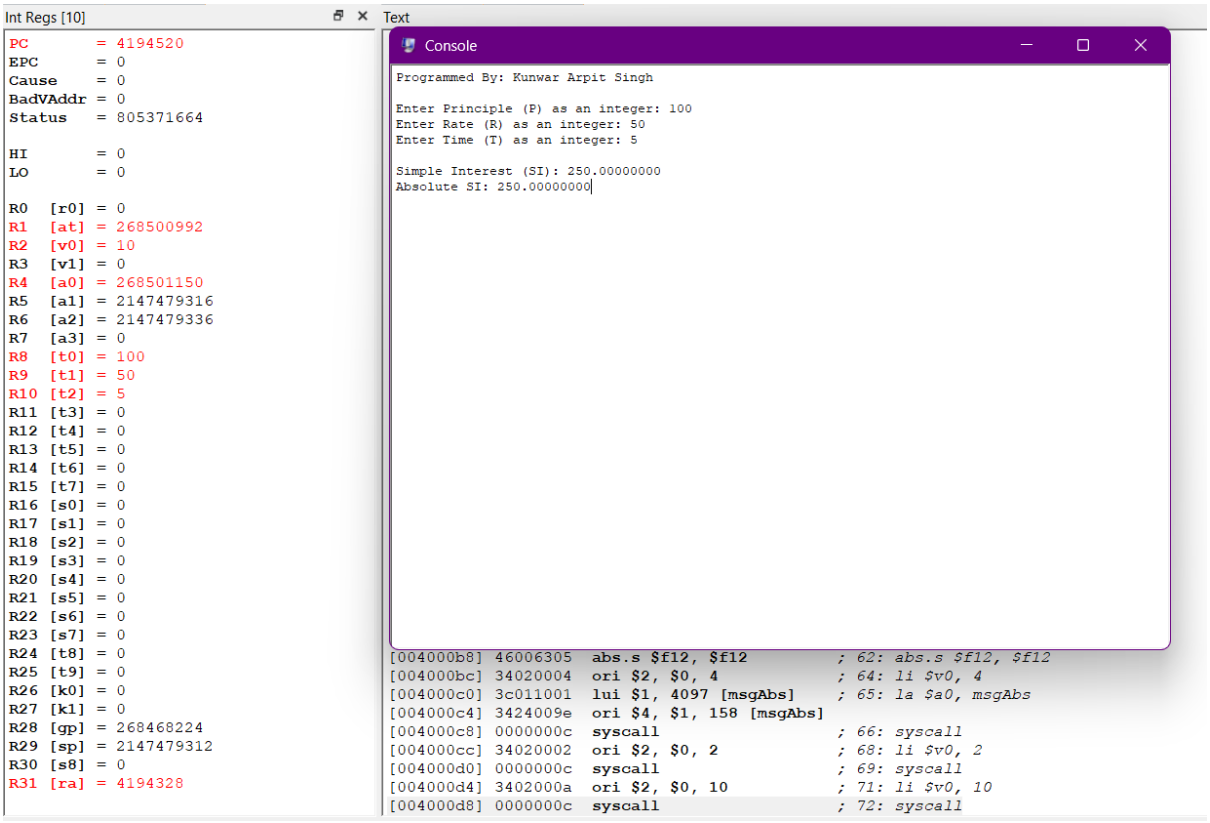
MIPS Assembly Source

```
# Assignment 6: Sequential Construct-II
# Programmed By : Kunwar Arpit Singh
.data
    promptP: .asciiz "Programmed␣By:␣Kunwar␣Arpit␣Singh␣\n\nEnter␣Principle␣(P)
        ␣as␣an␣integer:␣"
    promptR: .asciiz "Enter␣Rate␣(R)␣as␣an␣integer:␣"
    promptT: .asciiz "Enter␣Time␣(T)␣as␣an␣integer:␣"
    msgSI: .asciiz "\nSimple␣Interest␣(SI):␣"
    msgAbs: .asciiz "\nAbsolute␣SI:␣"
.text
.globl main
main:
    li $v0, 4
    la $a0, promptP
    syscall
    li $v0, 5
    syscall
    move $t0, $v0

    li $v0, 4
    la $a0, promptR
    syscall
    li $v0, 5
    syscall
    move $t1, $v0

    li $v0, 4
    la $a0, promptT
    syscall
    li $v0, 5
    syscall
    move $t2, $v0

    mtc1 $t0, $f1
    cvt.s.w $f1, $f1
    mtc1 $t1, $f2
    cvt.s.w $f2, $f2
    mtc1 $t2, $f3
    cvt.s.w $f3, $f3
    li.s $f10, 100.0
    mul.s $f4, $f1, $f2
    mul.s $f4, $f4, $f3
    div.s $f12, $f4, $f10
```

```
43
44      li $v0, 4
45      la $a0, msgSI
46      syscall
47
48      li $v0, 2
49      syscall
50
51      abs.s $f12, $f12
52
53      li $v0, 4
54      la $a0, msgAbs
55      syscall
56
57      li $v0, 2
58      syscall
59
60      li $v0, 10
61      syscall
```

Output Screenshots

Register values and Console



3. Problem 3

Problem statement

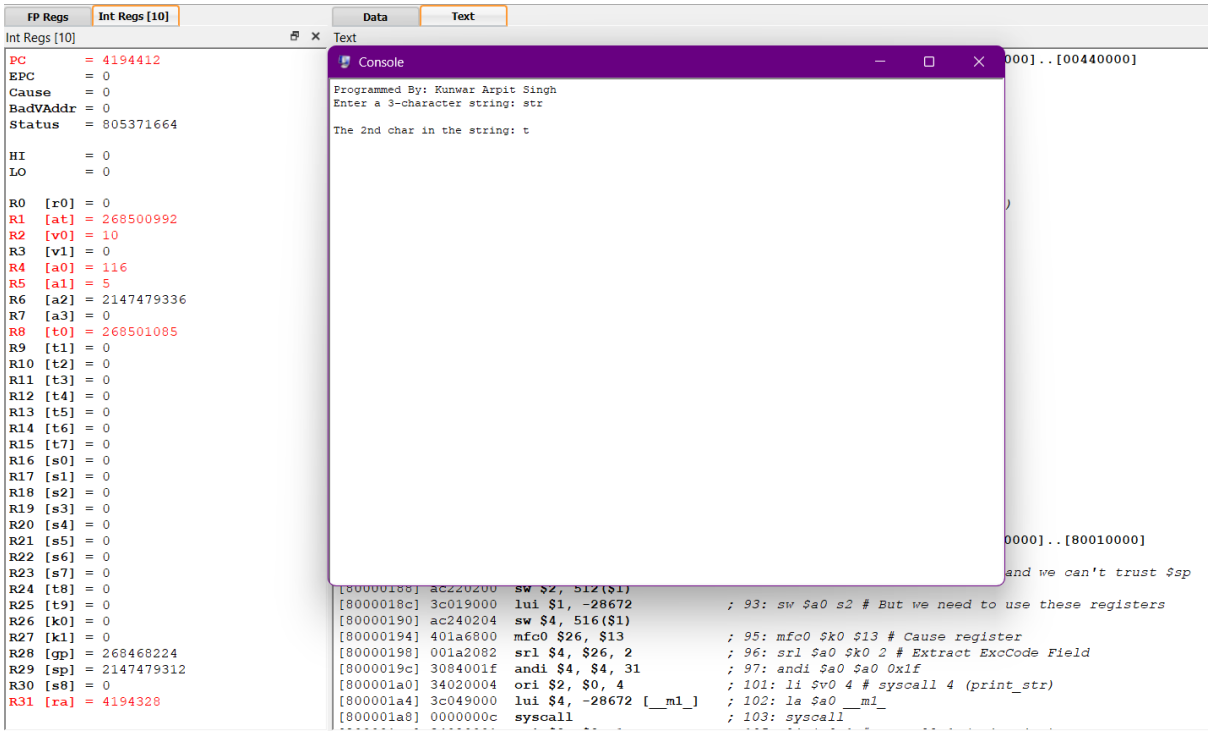
Write an assembly program that takes three character string from the user and print the second character from it. (Hint: use lbu instruction).

MIPS Assembly Source

```
# Assignment 6: Sequential Construct-II
# Problem 3
# Programmed By : Kunwar Arpit Singh

.data
    prompt: .asciiz "Programmed␣By:␣Kunwar␣Arpit␣Singh\nEnter␣a␣3-character␣
        string:␣"
    message: .asciiz "\nThe␣2nd␣char␣in␣the␣string:␣"
    buffer: .space 5

.text
.globl main
main:
    li $v0, 4
    la $a0, prompt
    syscall

    li $v0, 8
    la $a0, buffer
    li $a1, 5
    syscall

    li $v0, 4
    la $a0, message
    syscall

    la $t0, buffer
    lbu $a0, 1($t0)

    li $v0, 11
    syscall

    li $v0, 10
    syscall
```

Output Screenshots

Register values and Console



4. Problem 4

Problem statement

Write an assembly program that takes two strings (of length of two characters) from the user and calculate the hamming distance. For example if hi and he are the input from the user then the hamming distance is 01.

MIPS Assembly Source

```
# Assignment 1: Sequential Construct-I
# Programmed By : Kunwar Arpit Singh

.data
    prompt1: .asciiz "Programmed␣By:␣Kunwar␣Arpit␣Singh␣22185\n\nEnter␣first␣2-
        char␣string:␣"
    prompt2: .asciiz "\nEnter␣second␣2-char␣string:␣"
    result: .asciiz "\nHamming␣distance:␣"

    buffer1: .space 3
    buffer2: .space 3

.text
.globl main
main:

    li $t0, 0

    li $v0, 4
    la $a0, prompt1
    syscall
    li $v0, 8
    la $a0, buffer1
    li $a1, 3
    syscall

    li $v0, 4
    la $a0, prompt2
    syscall

    li $v0, 8
    la $a0, buffer2
    li $a1, 3
    syscall


    la $t1, buffer1
    la $t2, buffer2

    lbu $t3, 0($t1)
    lbu $t4, 0($t2)
```

```
41
42      sne $t5, $t3, $t4
43      add $t0, $t0, $t5
44
45      lbu $t3, 1($t1)
46      lbu $t4, 1($t2)
47
48      sne $t5, $t3, $t4
49      add $t0, $t0, $t5
50
51      li $v0, 4
52      la $a0, result
53      syscall
54
55      li $v0, 1
56      move $a0, $t0
57      syscall
58
59      li $v0, 10
60      syscall
```

Output Screenshots

Register values and Console